Supplementary Material for MorphJEPA: Morphology-Aware Latent Prediction for Hyperspectral Images

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1 Contents of the Supplementary Material

The supplementary material provides additional details and theoretical analysis supporting the main manuscript:

Sec. 1.1. Brief introduction to morphological operators and Joint-Embedding Predictive Architectures (JEPA).
Sec. 1.2. Mathematical formulation of the context-anchored Sketched Isotropic Gaussian Regularizer (SIGReg).
Sec. 1.3. Formal proofs for MorphJEPA's generalization and low-shot propositions.
Sec. 1.4 – Sec. 1.5. Dataset specifications and detailed evaluation protocols.
Sec. 1.6. Additional experiments including ablations on the number of PCA components, spatial patch size, and latent predictor depth, alongside a computational complexity comparison.

1.1 Brief Preliminaries: Morphological Operators and JEPA

Morphological Operators We use 2D mathematical morphology as a *structure preserving* operator on hyperspectral imagery (HSI). Let $\mathcal{X} \in \mathbb{R}^{H\times W\times B}$ denote an HSI scene defined on a discrete spatial domain $\Omega \subseteq \mathbb{Z}^2$ with B spectral bands. We treat morphology within the framework of complete lattices, where operators are defined based on the ordering of pixel values [9].

Given a spatial structuring element (SE) $S \subset \mathbb{Z}^2$, the two fundamental operators—*erosion* ε_S and *dilation* δ_S—are defined for a single spectral band $f : \Omega \to \mathbb{R}$ as:

$$\varepsilon_S(f)(p) = \inf_{z\in S_p} f(z), \quad \delta_S(f)(p) = \sup_{z\in S_p} f(z), \tag{1}$$

where $S_p = \{p + s \mid s \in S\}$ represents the translation of the SE to pixel p. Erosion expands dark regions (valleys), while dilation expands bright regions (peaks) [9].

Based on these primitives, we define the *morphological opening* γ_S and *closing* ϕ_S as:

$$\gamma_S(f) = \delta_S \circ \varepsilon_S(f), \quad \phi_S(f) = \varepsilon_S \circ \delta_S(f). \tag{2}$$

Crucially for our predictive framework, both opening and closing are *idempotent* filters, meaning $\gamma_S(\gamma_S(f)) = \gamma_S(f)$ and $\phi_S(\phi_S(f)) = \phi_S(f)$. Opening removes bright structures smaller than S (smoothing the foreground), while closing fills dark gaps smaller than S (smoothing the background) [3].

In MorphJEPA, these operations are applied band-wise across all B bands to generate the scene-level targets $\mathcal{X}^{(o)}$ and $\mathcal{X}^{(c)}$. This ensures that spatial boundaries and macro-region topologies are emphasized without artificially mixing spectral signatures. From a representation learning perspective, these operators act as deterministic, non-linear structural bottlenecks that selectively attenuate high-frequency spatial fluctuations while preserving the predictable topological skeleton of the semantic entities.

Joint-Embedding Predictive Architectures (JEPA) JEPA-style self supervised learning (SSL) frameworks learn representations by predicting the latent embedding of a *target view* from a *context view* in representation space [1]. Unlike generative methods, JEPA avoids pixel-level reconstruction; unlike contrastive methods, it avoids negative sampling.

Standard Formulation. Let $x \in \mathcal{X}$ denote an input sample. Stochastic transformations $\mathcal{T}_c, \mathcal{T}_t$ produce a *context view* $x_c = \mathcal{T}_c(x)$ and a *target view* $x_t = \mathcal{T}_t(x)$. A JEPA model consists of a context encoder $f_\theta : \mathcal{X} \to \mathbb{R}^D$, a target encoder $f_{\theta'} : \mathcal{X} \to \mathbb{R}^D$, and a predictor $h_\phi : \mathbb{R}^D \to \mathbb{R}^D$. The objective is to minimize the prediction error in latent space:

$$\mathcal{L}_{\text{JEPA}} = \|h_\phi(f_\theta(x_c)) - \text{sg}(f_{\theta'}(x_t))\|_2^2, \tag{3}$$

where $\text{sg}(\cdot)$ is the stop-gradient operator. To prevent representational collapse (where f_θ outputs a constant), standard JEPA updates the target encoder parameters θ' via an exponential moving average (EMA) of θ:

$$\theta' \leftarrow \tau\theta' + (1 - \tau)\theta, \tag{4}$$

where $\tau \in [0, 1)$ is a momentum coefficient.

LeJEPA Variant. LeJEPA [2] is a recent extension of JEPA that removes the need for both the momentum-based target encoder ($f_{\theta'}$) and the predictor network (h_ϕ). Instead of an asymmetric prediction task, LeJEPA stabilizes training by imposing an explicit *distributional regularization* symmetrically on the embeddings of all multi-views (local and global) in latent space. The objective is modified to force the empirical distribution of $z = f_\theta(x)$ to approximate a fixed reference distribution (typically an isotropic Gaussian, $z \sim \mathcal{N}(0, I)$). This symmetric regularization enforces global structure in latent space and enables stable training without momentum updates, predictors, or contrastive negatives, providing the foundational contrast to our context-anchored (asymmetric) regularization approach.

1.2 Latent Manifold Regularization (SIGReg Formulation)

As introduced in **Sec. 4.3** of the main paper, we apply the *Sketched Isotropic Gaussian Regularization* (SIGReg) [2] exclusively to the student context embed-

dings. Let $Z_c = \{z_c^{(k)}\}_{k=1}^{M} \subset \mathbb{R}^D$ denote a batch of M student context embeddings. The objective is to regularize their empirical distribution toward a target isotropic Gaussian prior $\mathcal{N}(0, I)$.

SIGReg relies on *characteristic function alignment* (CFA), since the characteristic function uniquely determines a probability distribution. To efficiently handle the high-dimensional latent space, the method employs 1D random projections (sketching). For a random projection direction $\mathbf{r} \sim \mathcal{U}(\mathbb{S}^{D-1})$, the projected scalar for a given embedding is defined as $u^{(k)} = \langle z_c^{(k)}, \mathbf{r} \rangle$. The *empirical characteristic function* (ECF) of the batch evaluated at frequency $t \in \mathbb{R}$ along direction $\mathbf{r}$ is:

$$\hat{\varphi}_{\mathbf{r}}(t) = \frac{1}{M} \sum_{k=1}^{M} e^{it\langle z_c^{(k)}, \mathbf{r} \rangle}. \tag{5}$$

For a standard normal distribution, the theoretical characteristic function is:

$$\varphi_{\mathcal{N}}(t) = e^{-t^2/2}. \tag{6}$$

The overall SIGReg penalty is formulated as a sketched *Epps–Pulley* test statistic. It minimizes the integrated squared discrepancy between the empirical and theoretical characteristic functions across all random projections:

$$\mathcal{L}_{\text{SIGReg}}(Z_c) = \mathbb{E}_{\mathbf{r}} \left[\int_{\mathbb{R}} \left| \hat{\varphi}_{\mathbf{r}}(t) - \varphi_{\mathcal{N}}(t) \right|^2 w(t) dt \right], \tag{7}$$

where $w(t) = e^{-\sigma t^2}$ acts as a Gaussian weighting function ensuring numerical stability.

In practice, the expectation $\mathbb{E}_{\mathbf{r}}$ is approximated via Monte Carlo sampling using P random projection vectors, and the integral over t is evaluated via numerical quadrature. By anchoring Z_c to this predefined prior, the regularizer explicitly enforces a bounded trace and near-isotropic covariance (as bounded in Proposition 2), ensuring a well-conditioned representation space.

1.3 MorphJEPA Generalization and Low-shot Benefits: A Simple Bound

Proof of Proposition 1: Representation Stability Under Nuisance Shift As established in **Sec. 4.5** of the main text, MorphJEPA's robustness under scene shifts can be formalized by analyzing the representation drift in latent space. Let an input patch be decomposed as $x = (s, n)$, where s denotes the class-relevant structural factor (morphology/topology) and n represents high-frequency nuisances (e.g., sensor noise, fine textures). Consider a target scene shift that perturbs the nuisance distribution while preserving the underlying structure, modeled as $\mathcal{P}'(s, n) = \mathcal{P}(s)\mathcal{P}'(n \mid s)$.

Following standard optimal transport bounds [7,8], the target risk $\mathcal{R}_{\mathcal{P}'}$ for a downstream predictor φ with an L-Lipschitz loss is bounded by the source risk and the Wasserstein-1 representation drift:

$$\mathcal{R}_{\mathcal{P}'}(\varphi \circ f_\theta) \leq \mathcal{R}_{\mathcal{P}}(\varphi \circ f_\theta) + L \cdot W_1(\mathcal{P}_z, \mathcal{P}'_z), \tag{8}$$

where $\mathcal{P}_z$ and $\mathcal{P}'_z$ are the pushforward distributions of the representation $z = f_\theta(x)$. To prove Proposition 1, we must show that predicting morphologically filtered targets actively contracts this W_1 distance.

We formalize this under three mild regularity assumptions. First, we assume **morphological attenuation** (A1). For $m \in \{o, c\}$, the morphologically filtered input suppresses nuisance variance such that $x^{(m)} = g_m(s) + r_m(n)$, where the residual mapping r_m is α-Lipschitz with $\alpha < 1$:

$$\|r_m(n_1) - r_m(n_2)\| \leq \alpha \|n_1 - n_2\|. \tag{9}$$

The degree of contraction in Proposition 1 scales with $(1 - \alpha)$; empirical verification that $\alpha < 1$ holds in practice is provided following the proof below.

Second, we assume **encoder Lipschitz continuity** (A2), meaning both the student encoder f_θ and the EMA teacher encoder $f_{\theta'}$ are K-Lipschitz:

$$\|f_{\theta'}(x_1) - f_{\theta'}(x_2)\| \leq K \|x_1 - x_2\|. \tag{10}$$

Third, we assume **predictive alignment** (A3) is achieved during pretraining, yielding approximate latent consistency between the student's context embedding and the teacher's morphological targets up to a residual prediction error ε:

$$\|f_\theta(x) - f_{\theta'}(x^{(m)})\| \leq \varepsilon. \tag{11}$$

Proposition 1 (Approximate contraction). Under assumptions (A1)–(A3), for any fixed structural factor s, the conditional representation drift satisfies:

$$W_1(\mathcal{P}(z \mid s), \mathcal{P}'(z \mid s)) \leq K\alpha W_1(\mathcal{P}(n \mid s), \mathcal{P}'(n \mid s)) + 2\varepsilon. \tag{12}$$

Proof. Let $x_1 = (s, n_1)$ and $x_2 = (s, n_2)$ be two inputs sharing the identical underlying structure s but differing in their nuisance components n. By (A1), their morphological targets are $x_1^{(m)} = g_m(s) + r_m(n_1)$ and $x_2^{(m)} = g_m(s) + r_m(n_2)$. Taking their difference perfectly cancels the invariant structural component $g_m(s)$:

$$x_1^{(m)} - x_2^{(m)} = r_m(n_1) - r_m(n_2). \tag{13}$$

By adding and subtracting the latent morphological targets from the teacher encoder, $f_{\theta'}(x_1^{(m)})$ and $f_{\theta'}(x_2^{(m)})$, the distance between the student representations can be bounded via the triangle inequality:

$$\begin{aligned}\|f_\theta(x_1) - f_\theta(x_2)\| \leq \|f_\theta(x_1) - f_{\theta'}(x_1^{(m)})\| + \|f_{\theta'}(x_1^{(m)}) - f_{\theta'}(x_2^{(m)})\| \\ + \|f_{\theta'}(x_2^{(m)}) - f_\theta(x_2)\|.\end{aligned} \tag{14}$$

Applying the predictive alignment bound (A3) to the first and third terms, and the Lipschitz continuity of the teacher (A2) to the middle term, we obtain:

$$\|f_\theta(x_1) - f_\theta(x_2)\| \leq \varepsilon + K\|x_1^{(m)} - x_2^{(m)}\| + \varepsilon = 2\varepsilon + K\|r_m(n_1) - r_m(n_2)\|. \tag{15}$$

Substituting the morphological attenuation bound from (A1) yields:

$$\|f_\theta(x_1) - f_\theta(x_2)\| \leq 2\varepsilon + K\alpha\|n_1 - n_2\|. \tag{16}$$

Now, fixing s and defining the conditional mapping $T_s(n) := f_\theta(s, n)$, the inequality above demonstrates that T_s is approximately $K\alpha$-Lipschitz:

$$\|T_s(n_1) - T_s(n_2)\| \leq K\alpha\|n_1 - n_2\| + 2\varepsilon. \tag{17}$$

Recall that the Wasserstein-1 distance between two distributions μ and ν is defined as $W_1(\mu, \nu) = \inf_{\pi \in \Pi(\mu,\nu)} \mathbb{E}_{(x,y)\sim\pi}[\|x - y\|]$. Integrating the point-wise inequality above over the optimal coupling π^* between $\mathcal{P}(n \mid s)$ and $\mathcal{P}'(n \mid s)$ yields:

$$W_1(T_{s\#}\mathcal{P}(n \mid s), T_{s\#}\mathcal{P}'(n \mid s)) \leq K\alpha W_1(\mathcal{P}(n \mid s), \mathcal{P}'(n \mid s)) + 2\varepsilon. \tag{18}$$

Since the pushforward measure $T_{s\#}\mathcal{P}(n \mid s) = \mathcal{P}(z \mid s)$, the result follows. This formally proves that predicting morphologically filtered targets contracts the conditional representation drift whenever $\alpha < 1$, with the contraction factor $(1 - K\alpha)$ scaling directly with the degree of nuisance attenuation. Empirical verification below confirms $\hat{\alpha} \approx 0.46$ in practice, yielding a mean contraction of $\approx 54\%$ per unit of nuisance drift and confirming the bound is non-vacuous.

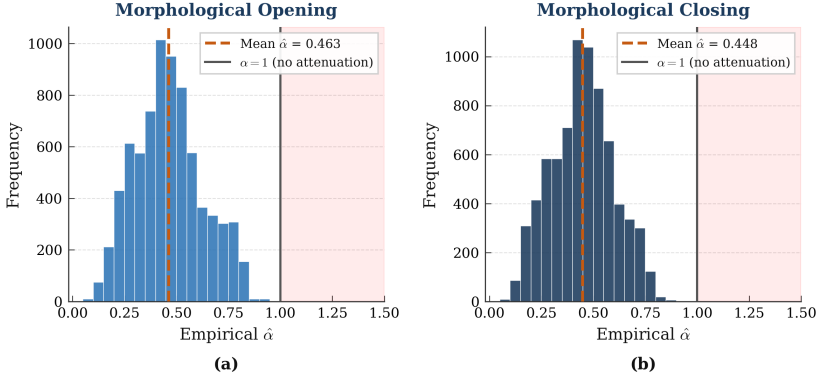


Fig. 1: Distribution of empirical attenuation factor $\hat{\alpha}$ for morphological (a) opening and (b) closing on Houston 2013. All patch pairs satisfy $\hat{\alpha} < 1$ (black vertical line), with zero violations (empty red region). Mean $\hat{\alpha} \approx 0.46$ (orange dashed line) indicates $\approx 54\%$ nuisance attenuation on average, confirming Assumption A1 is non-vacuous in our experimental setting.

Empirical Verification of Assumption A1. Proposition 1 requires the residual mapping r_m to be α-Lipschitz with $\alpha < 1$. We verify this empirically by estimating $\hat{\alpha}$ at the patch level on the Houston 2013 dataset. For same-class patch

pairs (x_1, x_2) — which share structural factor s by definition, isolating nuisance variation — we compute:

$$\hat{\alpha} = \frac{\|x_1^{(m)} - x_2^{(m)}\|_2}{\|x_1 - x_2\|_2}, \tag{19}$$

across 500 Monte Carlo pairs per class. As shown in Fig. 1, $\hat{\alpha} < 1$ holds for 100% of patch pairs under both operators, with mean $\hat{\alpha} \approx 0.46$ and 95th percentile of 0.77 (opening) and 0.71 (closing). This confirms Assumption A1 is satisfied in practice, with morphological filtering reducing nuisance variation by $\approx 54\%$ on average — making the contraction bound in Proposition 1 non-vacuous.

Proof of Proposition 2: Low-Shot Complexity and Covariance Control In **Sec. 4.5** of the main text, Proposition 2 establishes that context-anchored SIGReg improves the conditioning of the downstream linear probing task, tightening the sample complexity bound. Here, we provide the formal derivation bridging standard statistical learning theory to our specific representation constraint, demonstrating how MorphJEPA prevents the unbounded trace inflation typically caused by hyperspectral noise.

After self-supervised pretraining, we freeze the encoder and fit a linear classifier $c_\psi(z) = Wz$ using only N labeled samples. We consider a hypothesis class of linear predictors $\mathcal{F} = \{z \mapsto w^\top z \mid \|w\|_2 \le B\}$, where B is the weight norm bound. Let $\mathcal{Z} = \{z_1, \dots, z_N\}$ be a sample of N embeddings. The generalization gap is governed by the empirical Rademacher complexity of this class over the representation space [5]:

$$\widehat{\mathcal{R}}_{\mathcal{Z}}(\mathcal{F}) = \frac{1}{N}\mathbb{E}_\sigma\left[\sup_{\|w\|_2 \le B} \sum_{i=1}^{N} \sigma_i (w^\top z_i)\right], \tag{20}$$

where $\sigma_i \in \{-1, 1\}$ are independent Rademacher variables. By applying the Cauchy-Schwarz inequality, we can bound this supremum:

$$\widehat{\mathcal{R}}_{\mathcal{Z}}(\mathcal{F}) \le \frac{B}{N}\mathbb{E}_\sigma\left\|\sum_{i=1}^{N} \sigma_i z_i\right\|_2. \tag{21}$$

Using Jensen's inequality and the independence of the Rademacher variables ($\mathbb{E}[\sigma_i \sigma_j] = 0$ for $i \neq j$), we obtain the empirical bound:

$$\widehat{\mathcal{R}}_{\mathcal{Z}}(\mathcal{F}) \le \frac{B}{N}\sqrt{\sum_{i=1}^{N} \|z_i\|_2^2}. \tag{22}$$

Taking the expectation over the data distribution $\mathcal{Z}$ and applying Jensen's inequality once more yields the expected Rademacher complexity:

$$\mathcal{R}(\mathcal{F}) = \mathbb{E}_{\mathcal{Z}}\left[\widehat{\mathcal{R}}_{\mathcal{Z}}(\mathcal{F})\right] \le \frac{B}{N}\sqrt{\sum_{i=1}^{N} \mathbb{E}[\|z_i\|_2^2]} = B\sqrt{\frac{\mathbb{E}[\|z\|_2^2]}{N}}. \tag{23}$$

The expected squared norm of the representation is directly related to its covariance matrix $\Sigma = \mathbb{E}[(z - \mathbb{E}[z])(z - \mathbb{E}[z])^\top]$ via the trace operator:

$$\mathbb{E}[\|z\|_2^2] = \mathrm{Tr}(\Sigma) + \|\mathbb{E}[z]\|_2^2. \tag{24}$$

Our proposed context-anchored SIGReg plays a critical role here. By penalizing empirical characteristic function discrepancies strictly on the student network, SIGReg explicitly drives the marginal distribution of the context embeddings toward a centered isotropic Gaussian $\mathcal{N}(0, I)$. This centering effect successfully drives the mean toward zero ($\mathbb{E}[z] \approx 0$), simplifying the relationship to:

$$\mathbb{E}[\|z\|_2^2] \approx \mathrm{Tr}(\Sigma). \tag{25}$$

Because the teacher network is an EMA of the student, this well-conditioned manifold structure naturally propagates to the target representations without requiring explicit regularization on the freely trainable targets themselves. This synergy encourages the downstream covariance to approximate the identity matrix ($\Sigma \approx I$). In finite-sample and empirical optimization regimes, this holds up to a small residual covariance error $\|\Sigma - I\|_2 \leq \epsilon$ for some optimization residual $\epsilon > 0$.

Consequently, the eigenvalues of Σ are bounded by $1 + \epsilon$, which restricts the trace:

$$\mathbb{E}[\|z\|_2^2] \approx \mathrm{Tr}(\Sigma) \leq \mathrm{Tr}((1 + \epsilon)I) = D + D\epsilon = D + \delta, \tag{26}$$

where $\delta = D\epsilon$ represents the residual trace error. Substituting this trace bound back into the expected Rademacher complexity yields the final analytical bound:

$$\mathcal{R}(\mathcal{F}) \leq \mathcal{O}\left(B\sqrt{\frac{D + \delta}{N}}\right). \tag{27}$$

In summary, compared to standard JEPA, MorphJEPA optimizes both core terms of the generalization bounds. The morphological prediction objective filters out nuisance variance, reducing the representation drift W_1 under scene shift (as proved earlier). Simultaneously, the context-anchored SIGReg objective, propagated via the EMA teacher, forces a full-rank, isotropic covariance ($\Sigma \approx I$). This prevents any single nuisance direction from inflating $\mathrm{Tr}(\Sigma)$ [4] while avoiding the semantic collapse inherent to regularizing targets symmetrically. Together, these mechanisms restrict the hypothesis complexity to scale predominantly with the chosen latent dimension D, allowing optimal linear hyperplanes to be found reliably even in extreme ($N \in \{1, 5, 10\}$) low-shot regimes.

1.4 Dataset Details

To evaluate the representation learning and transfer capabilities of MorphJEPA, we conduct experiments on standard hyperspectral benchmarks covering both intra-scene and cross-scene settings.

Houston 2013 (Intra-Scene) [11]. Acquired using the CASI-1500 sensor, this dataset captures an urban environment around the University of Houston campus. The hyperspectral cube spans the spectral range 0.38–1.05 µm with 144 spectral bands. The scene has a spatial resolution of 2.5 m per pixel and dimensions of 340×1905 pixels. The ground truth contains 15,029 labeled samples across 15 land-cover classes.

Trento (Intra-Scene) [11]. This dataset captures a rural area in Trento, Italy, and was acquired using the AISA Eagle sensor. It contains 63 spectral bands covering 0.42–0.99 µm with a spectral resolution of 9.2 nm and a spatial resolution of 1.0 m per pixel. The image size is 166×600 pixels with 6 semantic classes and 30,214 labeled samples.

Houston 2013–2018 (Cross-Scene) [13]. To evaluate temporal cross-scene transfer, we utilize the Houston benchmark consisting of scenes captured in 2013 and 2018. These scenes cover the same geographical region but were acquired using different sensors and spatial resolutions. The 2013 scene contains 144 bands at 2.5 m resolution, while the 2018 scene provides 48 bands at 1.0 m resolution. For cross-scene alignment, 48 overlapping spectral bands from the 2013 scene are retained and a corresponding spatial region of 209×955 pixels is extracted. Both scenes share 7 consistent land-cover classes.

HyRANK (Cross-Scene) [13]. HyRANK is a satellite-based benchmark acquired by the Hyperion sensor (EO-1, USGS) containing 176 spectral bands. We evaluate cross-scene transfer between two geographically distinct scenes: Dioni (250×1376 pixels) and Loukia (249×945 pixels). Both scenes share 12 consistent land-cover classes.

1.5 Detailed Pretraining and Evaluation Protocol

Due to space limitations in the main manuscript, we provide additional details regarding the evaluation setup introduced in **Sec. 5.1**. Hyperspectral benchmarks consist of continuous spatial–spectral cubes with sparsely distributed labeled pixels. In this setting, the scene constitutes a single spatial–spectral observation, while labeled pixels act as sparse annotations within that observation.

Intra-Scene Evaluation. MorphJEPA is pretrained in a purely self-supervised manner, adhering to standard practices for continuous HSI representation learning [6,10,12,14]. During pretraining, anchor patch centers are sampled exclusively from non-test spatial locations $\Omega \setminus \mathcal{T}$ (comprising unlabeled and training pixels). For example, in the Trento scene ($166 \times 600 = 99{,}600$ pixels with $\sim$30,000 test samples), this yields approximately 69,000 valid patch centers. While the extracted 15×15 contextual patches naturally encompass overlapping neighborhood pixels due to the continuous nature of the scene, the predictive objective remains strictly label-free, and no test pixels serve as prediction anchor centers. Following pretraining, the encoder is frozen. We randomly sample N labeled pixels per class (1, 5, and 10-shot) from the benchmark training set to fit a linear classifier. Performance is then evaluated on the standard benchmark test set.

Cross-Scene Generalization. For cross-scene evaluation, the *target* scene is completely unseen during pretraining. MorphJEPA is pretrained exclusively on the *source* scene. We consider two downstream settings:

1. **Zero-Shot Target Transfer:** N labeled samples per class are drawn from the source scene to train the linear classifier, and evaluation is performed directly on the target-scene test samples.
2. **Low-Shot Target Adaptation:** N labeled samples per class are drawn from the target scene to train the linear classifier, and evaluation is conducted on the remaining target samples.

Comparison Protocol for Baselines and SOTA Methods. To ensure a fair comparison, all standard SSL baselines and state-of-the-art (SOTA) HSI frameworks were evaluated under a unified protocol. For standard SSL baselines (SimCLR, MAE, I-JEPA, and LeJEPA), we utilized MorphJEPA's disentangled spatial–spectral encoder as the shared backbone. Generic RGB backbones cannot process hyperspectral inputs without conflating spectral mixing and spatial aggregation; thus, our encoder provides a necessary, domain-compatible foundation. This isolates the pretraining objective as the sole source of performance differences. MorphJEPA's backbone-agnostic nature is further corroborated in the main text (**Sec. 5.4, Tab. 5**), which demonstrates consistent gains across standard 3D-CNN and HSI-transformer architectures.

For existing HSI-specific SOTA methods, we retained their original architectures. To ensure these frameworks achieved their maximum potential, their learning hyperparameters were rigorously optimized to extract the best possible performance under our specific 1, 5, and 10-shot evaluation regimes. Furthermore, to guarantee a strictly equitable comparison across all frameworks, the input patch size and the number of retained PCA components were universally aligned.

1.6 Additional Experiments and Results

In this section, we present additional experiments evaluating key architectural choices, including the number of retained PCA components, spatial patch size, and predictor depth. We also compare the floating-point operations (FLOPs), parameter counts, and inference time of MorphJEPA with other SOTA HSI methods in Tab. 2.

PCA Components. We ablate the number of retained PCA components over $\{10, 15, 20, 25\}$, as shown in Fig. 2(a). Performance peaks at 15 components for intra-scene (Houston 2013) and both cross-scene low-shot target adaptation benchmarks (Houston, HyRank). Using fewer components discards discriminative spectral variance, while larger values ($\geq$20) introduce channels dominated by sensor noise. Applying morphological operators to such unstructured noisy components generates unstable spatial responses, which degrade the pretext learning signal. Although the structurally simpler Trento scene marginally favors $C = 20$ (89.85% vs. 89.62%), we fix $C = 15$ across all datasets to maintain a strict, dataset-agnostic evaluation protocol.

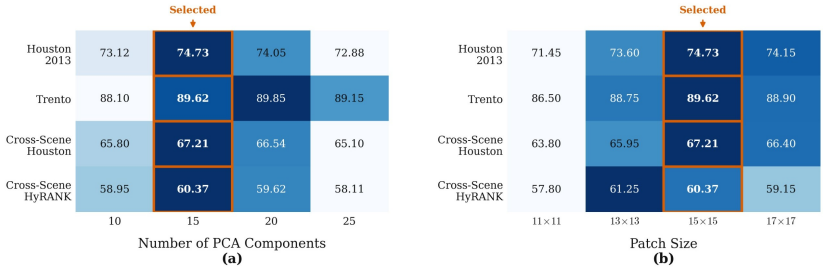


Fig. 2: Hyperparameter sensitivity across intra-scene and cross-scene benchmarks. 5-shot OA (%) is reported. **(a)** Impact of the number of retained PCA components. **(b)** Impact of the spatial patch size. The highlighted orange bounding boxes indicate the globally selected hyperparameters ($C = 15$ and 15×15 patch size) adopted to enforce a strict, dataset-agnostic evaluation protocol.

Spatial Patch Size. As shown in Fig. 2(b), we evaluate spatial windows ranging from 11×11 to 17×17 under both intra-scene and cross-scene (low-shot target adaptation) settings. A 15×15 window performs best on Houston 2013, Trento, and cross-scene Houston. Smaller patches provide insufficient spatial support for the 3×3 morphological operators to capture meaningful macro-topology, while larger patches (17×17) span multiple semantic regions, weakening local spectral consistency. Although HyRank slightly prefers 13×13 (61.25% vs. 60.37%), we adopt 15×15 uniformly to avoid dataset-specific tuning.

Latent Predictor Depth. Tab. 1 evaluates the effect of latent predictor depth $(h_{geo}, h_{open}, h_{close})$ under the 5-shot intra-scene regime. Performance peaks with a 2-layer MLP (74.73% OA on Houston 2013, 89.62% OA on Trento). A single linear layer underfits the context-target alignment morphological task (−1.88% OA on Houston), while deeper predictors (3–4 layers) degrade performance. Overly expressive predictors can implicitly model the morphological transformations, weakening the representation learning pressure on the encoder. A 2-layer MLP, therefore, provides an effective bottleneck that encourages the encoder to learn robust spatial–spectral features.

Table 1: Ablation of latent predictor depth under 5-shot regime on intra-scene (Houston 2013, Trento) datasets. 2-layer predictor provides the optimal structural bottleneck, preventing the predictor from acting as a computational crutch. Best results are highlighted in red.

Predictor Depth	Houston 2013			Trento		
	OA	AA	κ	OA	AA	κ
1-layer (Linear)	72.85	75.41	70.52	88.21	83.12	84.77
2-layer (MorphJEPA)	74.73	77.53	72.61	89.62	84.40	86.20
3-layer	73.12	76.15	71.08	88.93	83.78	85.31
4-layer	71.48	74.55	69.22	87.44	82.11	83.62

Table 2: Computational complexity comparison of MorphJEPA with SOTA HSI methods in terms of FLOPs, trainable parameters, and inference time. Best results are highlighted in red.

Method	FLOPs (G)	Params (M)	Inference (ms)
MorpMamba	0.03	0.07	1.15
Res-CP	0.63	0.28	1.48
DMVL	2.12	24.62	5.87
SC-EADNet	0.02	1.51	13.52
SSLSM	0.39	25.65	6.02
HSI-MAE	1.52	3.18	2.15
GGCL	0.05	0.19	1.08
SpectralDINO	0.75	21.55	12.16
CM-SCON	1.85	1.45	4.35
MorphJEPA	0.04	0.18	0.98

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